ESE 561 Why Robbins–Monro Works: a Guided Proof

problems 10

Setting. All our algorithms (bandit estimates, Monte Carlo, TD, SARSA, Q-learning) update an estimate by

$$x_{k+1} = x_k + \alpha_k\,(Y_k - x_k), \qquad k = 0, 1, 2, \ldots, \qquad \alpha_k \in [0, 1],$$

where the targets Y_k are independent, $\mathbb{E}Y_k = \mu$, $\operatorname{Var} Y_k = \sigma^2 < \infty$. **Claim** (Robbins and Monro, 1951): if $\sum_k \alpha_k = \infty$ and $\sum_k \alpha_k^2 < \infty$, then $x_k \to \mu$. Throughout, $e_k = x_k - \mu$ is the error and $\xi_k = Y_k - \mu$ the noise ($\mathbb{E}\xi_k = 0$, $\mathbb{E}\xi_k^2 = \sigma^2$).

Problem 1. *Error recursion.* Show that $e_{k+1} = (1 - \alpha_k)\,e_k + \alpha_k\,\xi_k$. Interpret the two terms.

Problem 2. *Unrolling.* Compute e_2 in terms of e_0, ξ_0, ξ_1; then prove by induction

$$e_n = \prod_{k<n} (1 - \alpha_k)\,e_0 + \sum_{k<n} \alpha_k \prod_{k<j<n} (1 - \alpha_j)\,\xi_k \qquad \text{(initial error + accumulated noise)}.$$

Problem 3. *The first condition forgets the start.* (a) Show $1 - \alpha \le e^{-\alpha}$ for all real α. (b) Deduce that the initial error in Problem 2 vanishes, $\prod_{k<n}(1 - \alpha_k) \to 0$, if $\sum_k \alpha_k = \infty$.

Problem 4. *Mean-square error.* Let $u_k = \mathbb{E}[e_k^2]$. Square the recursion of Problem 1 and show

$$u_{k+1} = (1-\alpha_k)^2 u_k + \alpha_k^2 \sigma^2 \;\le\; (1-\alpha_k)\, u_k + \alpha_k^2 \sigma^2.$$

Why does the cross term vanish? Where is $\alpha_k \le 1$ used?

Problem 5. *A lemma.* Let $u_{k+1} \le (1-\alpha_k) u_k + c_k$, with $0 \le \alpha_k \le 1$, $\sum \alpha_k = \infty$, $c_k \ge 0$, $\sum c_k < \infty$. Prove $u_k \to 0$. *Hint:* unroll as in Problem 2; for the sum, fix $\epsilon > 0$ and split it at an index K with $\sum_{k \ge K} c_k < \epsilon$.

Problem 6. *Putting it together.* Apply the lemma to Problem 4. Which condition is used where? Bound $P(|x_k - \mu| > \epsilon)$.